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Decision Analysis

Risk—Its Priority and Probability: The Analytic Hierarchy Process

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Risk estimation involves priorities and probabilities which are themselves a form of priority of natural alternatives. This paper provides illustrations of how one can deal with risk and uncertainty using the Analytic Hierarchy Process, a new approach to measurement by ratio scales. The paper also includes a discussion of how to deal with risk strategies involving interdependence. Particular emphasis is made on the siting of nuclear power plants.

KEY WORDS: Risk; Uncertainty; Decision Making; Hierarchies; Low Probability Events.

1. INTRODUCTION

A risk situation differs from benefit–cost analysis as it involves potentially very high unacceptable costs that no one expects to pay. Here, an individual or group put themselves in a position where things may not work out as expected, leading to losses far exceeding the costs of the undertaking itself. What are these situations and which ones are worth the risk? Why do some people find it natural to take a chance on risky endeavors while others do not? Risk analysis must, of necessity, involve different time horizons over which the stability of the system is investigated. It can also involve different geographic locations where the penalty can be minimized. Finally, in case things go wrong, quick action with appropriate systemic controls may be followed to minimize damage. Thus, risk analysis is not analogous to throwing dice in a game framework. Rather it is a complex real world setting that can be favorably managed with intelligence, creativity, and prior planning. These are the aspects that must be emphasized rather than the mathematical estimation of the probabilities of the things that can go wrong. Undertaking risky activity is not a simple matter of courage and blind commitment.

It has been known for a long time that we do not have accurate and justifiable methods to make predictions for the future based on the past. But, people blithely go on doing it because they do not know any better. At first, when people use it they recognize that it is not a sure way. The longer groups of people use it and give excuses for it, the more likely they are to become conditioned by their own habits and feel outraged if it is challenged, or feel even worse if it fails (as it often does). Instead of backing off to the weak original position, they entrench themselves in a defensive position. In the end, misguidedness prevails with flimsy justification. It may even be better to consult a medicine man or a soothsayer than to use this extrapolative form of fortune telling cloaked in the guise of science or statistics.

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The problem is the same with risk analysis. Dressing a skeleton with fancy clothes does not give it life. Here again, the field suffers from elaborate techniques purported to deal with risk and have fallen short in real life applications. Let us comment on the state of nature and our ability to know it.

A sufficient account of the possibility of disorder should be included in our representation of risk to capture what might happen. We should never be smug and cocksure. This kind of suspicion is likely to be helpful against surprises.

Another problem is that we are accustomed to dealing with complexity one factor at a time, in a chain like manner, when actually things are interdependent. We need a way to represent such interactions and dependencies holistically in our structures for handling risk.

A third assumption, particularly by those who have a good mind for analysis, is that we can get probabilities from our minds by pretending that we understand the situation and by doing reasoning analogous to things we really do understand (as if nature is a simple coin or a series of well-organized devices clearly interpretable in terms of probabilities). Still worse, often our probabilities are not determined experimentally because we cannot set up experiments; we lack the knowledge even to simulate them for catastrophic events that we think (or hope) have a low probability of occurrence. It is a process of self-hypnosis when we convince ourselves and others that we have the right probabilities because they appear to be so natural, given our sensibly sounding causal explanations. We should be especially wary of such explanations and our proclivity to estimate probabilities.

A final point is that an old Babylonian habit remains active in our technical minds. We think that the world is a riddle, and like magicians belonging to a brotherhood with a tradition and a language, we can unravel it if we can only find the key. Our modern key is to resort to complex theoretical manipulations, thinking that, if pursued with sufficient diligence, the right combination will fall in place and we would have the answer.

A case in point is the traditional approach to the disposition of an individual to risk, based on the concept of utility function and the rate of change of its second derivative with respect to the first derivative. The calculation of the utility function involves the probability of risk. People have complained that this kind of technique-driven approach, neat as it is, is hard to apply to their problems, and leans too heavily on theoretical manipulations.

Instead, we need a better understanding and representation of complexity and of the attitudes needed to estimate what goes on in that complexity, and different potential scenarios that could result from a combination of the structures we use and the attitudes we assume. Additionally, we need a judgment about the entire system of thinking that we use in this situation: whether it is destined to work, and what alternatives we have to deal with the problem.

2. PRIORITIES, PROBABILITIES AND UNCERTAINTY

Priority is the relative intensity of what is important to people. Probability or likelihood is nature's priority at the occurrence level. Note that complexity can be better represented through priority because there are usually elements whose contribution or influence is not probabilistic, but nevertheless describable in terms of priorities. For example, given a set of factors, one can ask which of them has greater influence on the outcome, or which one has the higher priority influence. This lack of complete representation has been a serious flaw in risk analysis where everything is described in terms of probabilities alone.

In modern risk analysis we need to determine numerically not just what is likely to happen, but also what is important and what is not important if it does happen. One must consider all observed factors and then establish priorities in the two senses we mentioned above: importance and likelihood of occurrence.

There are 2 types of uncertainty: (a) uncertainty about the occurrence of events, and (b) uncertainty about the range of judgments used to express preferences. The first is beyond the control of the decision maker. The second is a consequence of the amount of information available to him and his understanding of the problem.

We have studied[1] the second type of uncertainty experienced by the decision maker in making pairwise comparisons. We assume that one can associate each judgment with an interval of numerical values from which a reciprocal matrix of interval

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values in each position is constructed. The problem then is to generate the possible solutions from such a matrix by assuming various probability distributions. The approach has also been generalized to an entire hierarchy. The purpose of the analysis is to study the likelihood that alternatives change rank and, in particular, the likelihood that a high priority alternative could become lost due to uncertainty and change ranks with a lower priority one (and in bad cases, with a really low priority one). This approach has great relevance to problems of uncertainty and high risk.

3. A DECISION ON INVESTMENT

A recently retired person wanted to formulate an investment portfolio tailored to his particular needs. Most of us can identify readily with this individual's problem of grappling with risk, uncertainty, and other abstract intangibles. He constructed the following hierarchy (Fig. 1):

For simplicity, the second level contained only four criteria which are in turn an aggregation of many subcriteria that came to mind. The first criterion, return, includes interest, dividends, and capital appreciation. Although capital appreciation and short-term income are taxed differently, this difference is taken into account in another group. Estimated annualized returns were used when judging the individual securities. The next criterion, low risk, included subcriteria involving diversification and low-perceived risk and volatility of return. The third criterion, tax benefits, assumes that capital gains are taxed lower than short-term returns and that tax-free investments, such as municipal bonds, would be rated high. The fourth criterion, liquidity, included small transaction costs, ease of withdrawal, and the ability to make small additions. These second-level-clustered criteria could have been followed by a third level of subcriteria consisting of all the above itemized breakdowns, but, to be brief, we have gone directly to the portfolio aggregate alternatives in the next level.

The third level consists of the following alternatives: the first, tax free bonds, are low return, low risk securities with great tax advantage; the second, securities, are tied to the treasury rate with medium return, low risk, and relative liquidity; the third, saving accounts, involve low return, but their liquidity and very low risk are their advantage; the fourth, speculative stocks, involve high return and high risk investments and include the options market; and the fifth, blue chip stocks and bonds, are the medium return, low risk investments.

The judgment phase of the analytic hierarchy process (AHP) requires the following scale of absolute values (not ordinals) to express judgments in making paired comparisons: 1, equal; 3, moderate; 5, strong; 7, very strong; 9, extreme (2, 4, 6, 8 for compromise—reciprocals for the inverse comparison, and decimal refinements between if it is desired to obtain a predetermined set of final priorities). The smaller of two elements being compared is considered to be the unit and the larger one is assessed to be so many times more than it, using the intensity of feeling and translating it to the numerical value. One can show that the derived scale from the compari-

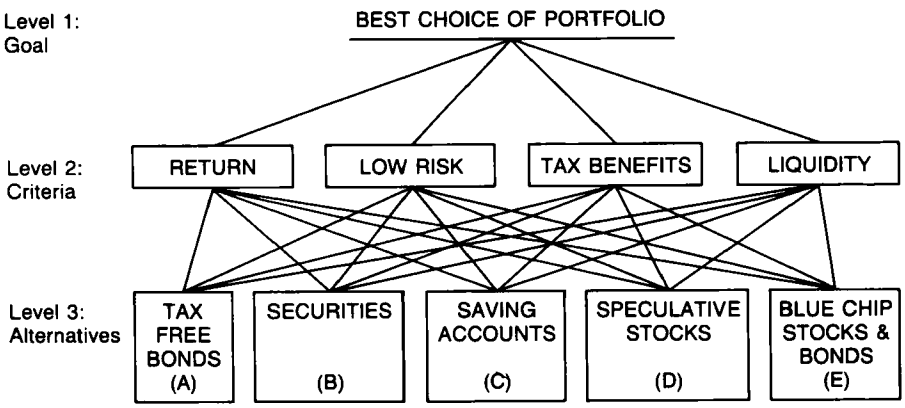

Fig. 1. Hierarchy for the best choice of a portfolio.

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sons is insensitive to small changes (corresponding to uncertainty between two neighboring intensities) in the values of the paired comparisons. If exact numbers for the comparisons are known, and their ratios are used, the derived scale gives their relative value back. However, even when exact numbers are available, one ordinarily needs to compare the value of these numbers by interpreting their relative importance and then using that instead of the numbers themselves which may have no universal meaning (e.g., to a family of modest income, a $20,000 car may be much more expensive than a $10,000 car, but to a military budget planner they may be about the same).

The next step after structuring the hierarchy is to carry out comparative judgments using the above scale. The elements in the second level are arranged in a matrix and judgments are elicited as to the relative importance of each criterion when compared with every other criterion on that level. The question here is: for a best choice of portfolio for the particular individual's circumstances (which give rise to his judgments), which criterion does he consider more important and how strongly more? We have the following matrix (Table I.) for comparing the criteria with respect to the goal, and the weights derived from it:

A criterion (X), represented on the left, is compared with respect to a criterion (Y), represented on the top of the matrix. If X is more important than Y, then a numerical value greater than 1 is used in the (X, Y) position. If Y is more important than X, then the reciprocal of this value is used. The reciprocal of whatever value is entered in the (X, Y) position is automatically entered in the (Y, X) position.

For example, in answering the question, "Which is more important, return on the left, or low risk on top?," we believe that low risk is strongly more important than return and, hence, enter the value 1/5 in the first row, second column position and automatically enter the value 5 in the second row, first column position, and so on.

The paired comparisons give rise to a ratio scale of weights of the relative importance or priorities of the criteria. Ratio scales are a strong class of numbers whose ratios remain the same when each of them is multiplied by a positive constant. For example, the ratio of the weight of 2 objects measured either in pounds or in kilograms is the same. Ratio scales make it possible not only to rank alternatives, but also to allocate resources appropriately in proportion to the values.

Table I.

	Return	Low risk	Tax benefits	Liquidity	Weights
Return	1	1/5	3	3	.214
Low risk	5	1	5	4	.598
Tax benefits	1/3	1/5	1	1	.090
Liquidity	1/3	1/4	1	1	.097

The AHP does not insist on consistent judgments and provides an index for measuring inconsistency, both for each matrix of comparisons and for the entire hierarchy. It is also possible to find out where the most inconsistent judgments are and to change them if desired to improve the consistency, although this is not required. The inconsistency ratio for the above matrix is 0.072 which is less than the tolerance level of 0.100. The reason for using a tolerance level of 0.100 is this: although the mind is primarily concerned with constructing a consistent decision, it must allow a modicum of inconsistency in order to admit new information, giving rise to change in the old judgments. However, inconsistency is less important than consistency by one order of magnitude (the 10% tolerance range)[2].

Next, the alternatives are compared in pairs with respect to each criterion in separate matrices. The resulting scales from the 4 matrices are each weighted or multiplied by the importance of the criteria just derived above, and then added to obtain the overall importance of the alternatives (Table II.). We have:

This portfolio was compared with the investor's existing portfolio (the percentages shown in the last column). Although at present the investor makes use of the second and third investments appropriately, the bulk of the portfolio is tied up in medium return, low risk corporate securities. He decided to sell a large part of these and reinvest in municipal bonds and more speculative issues to satisfy his investment objectives.

4. THE SITING OF A NUCLEAR POWER PLANT

The AHP provides a rational mechanism to guide the process of public policy decision towards a more responsible, consistent approach to the man-

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Table II.

	Return (.214)	Low risk (.598)	Tax benefits (.090)	Liquidity (.097)	Final priority	Existing portfolio
A	.054	.162	.521	.051	.16	.00
B	.165	.338	.057	.298	.27	.30
C	.057	.344	.057	.487	.27	.20
D	.497	.034	.124	.068	.15	.00
E	.227	.122	.241	.096	.15	.50
Inconsistency	.031	.031	.023	.025		
Composite inconsistency		.025				

agement of risk. In no instance is this need more acutely felt than in the case of a nuclear power plant siting decision, since it entails the full gamut of claims and counter-claims of an environmental, technological, social, and economic character.

Technological risk involves two components: the likelihood of an event occurring, and the nature of the distribution of its negative impact along various dimensions. With regard to such impacts, they include: the character of the potential loss (e.g., whether it is a monetary loss, a physical injury or death, or an environmental disability), the extent of the loss in terms of its intensity and diffusion, and its timing (how immediate is the danger, what would be its frequency, and so on).

The AHP can be used to address these negative elements within an overall cost–benefit framework. We would thus structure the nuclear power plant siting problem into two hierarchies, a costs hierarchy and a benefits hierarchy (see Figs. 2–6).

Priorities are established on the subjective evaluations (by the actors in levels 2 and 3 of Fig. 2) of the losses resulting from building the proposed nuclear power plant. Priorities are also set on the actor's objectives in level 4 of Fig. 2. To incorporate the kinds of losses involved, the cost factors are clustered according to whether they are economic, physical, or social (see level 5 of Fig. 2). In yet another level, each cluster is decomposed into an itemized list of subfactors (level 6 of Fig. 2). Again, the subfactors are decomposed according to uncertainty by estimating the degree of their intensity: high, medium, or low (level 7 of Fig. 2). Further, each of these is decomposed into "size of population affected" categories, ranging from large to medium to a small number of people (level 8 of Fig. 2).

Finally, the alternative locations and nuclear technologies that may be used are identified in a separate level and prioritized in terms of the intensity cost factors as they impact on different sizes of the population.

This is meant simply to serve as an example of how such a hierarchy would be structured. If one wished to determine the priorities of various institutional constraints (e.g., political and administrative considerations), the alternate sites and technologies can be followed by a still lower level of feasible implementation strategies. (For costs hierarchy, see Fig. 2).

Now another hierarchy of the "benefits" is developed. An essential difference in the two hierarchies would be the level 6 specific effects, which now lists the positive aspects of the technology. This level is as follows: specific desirable effects—cheap power, clean power, greater productivity, neat environment, high technology, prestige, equity.

The prioritization process of costs and benefits would then be used to form benefit–cost ratios. It has been our experience that such ratios are most meaningful when familiar technologies that have been adopted and used by society for some time are included to serve as comparison yardsticks. AHP could also easily be used to perform sensitivity analysis with regard to variations in the structure of the problem and in the judgments. It is also a means of explicitly measuring consistency of the judgments.

Figures 2 and 3 give the complete benefit–cost analysis representation of the problem of siting nuclear power plants.

In the next section, we give a generalization of the hierarchic approach to risk problems involving interdependence. The theory will be given without particular emphasis on risk but, based on the earlier applications, the reader will be able to see how it applies.

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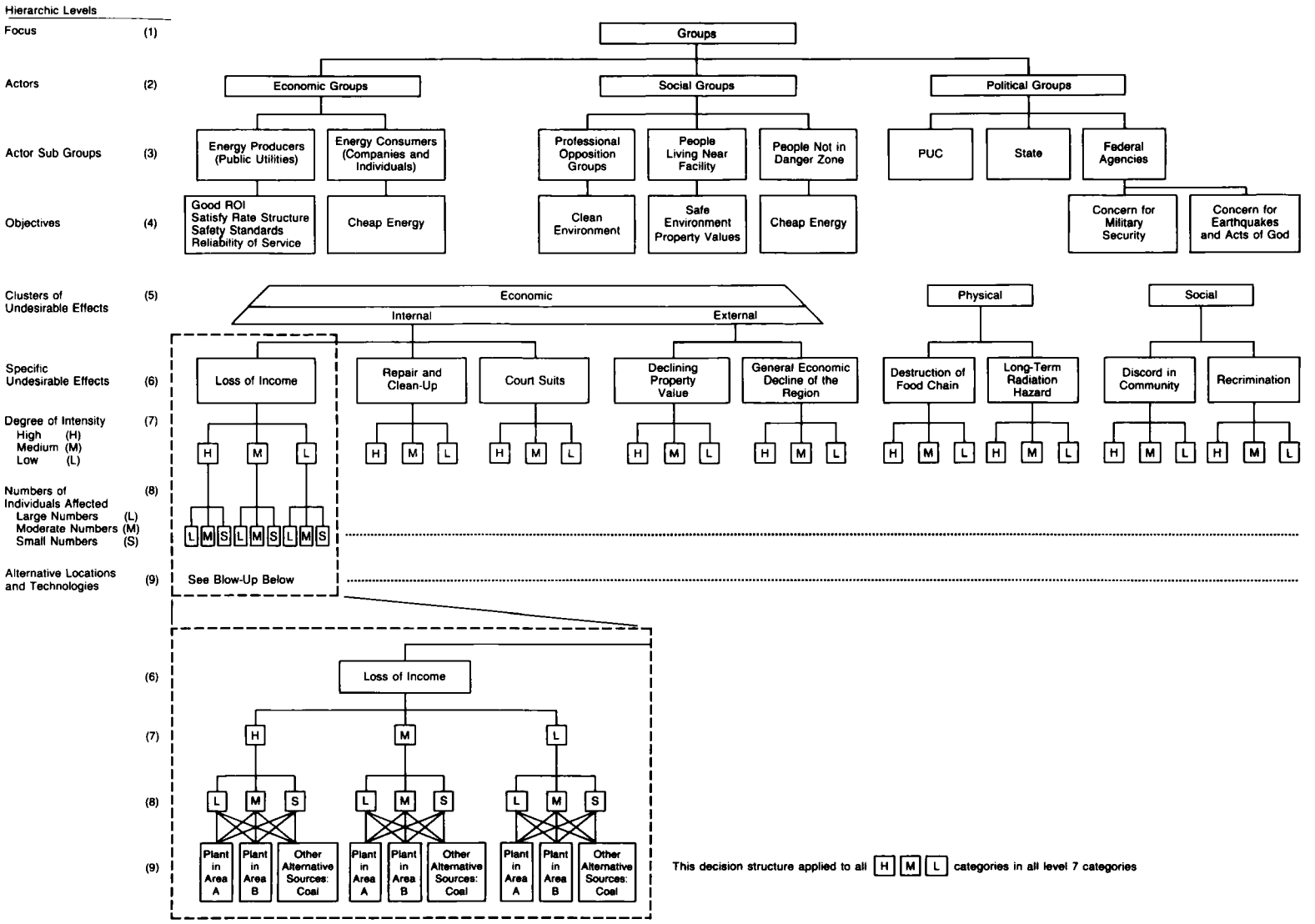

Fig. 2. Costs hierarchy for siting of a nuclear power facility. Levels 1–5 are common with Costs.

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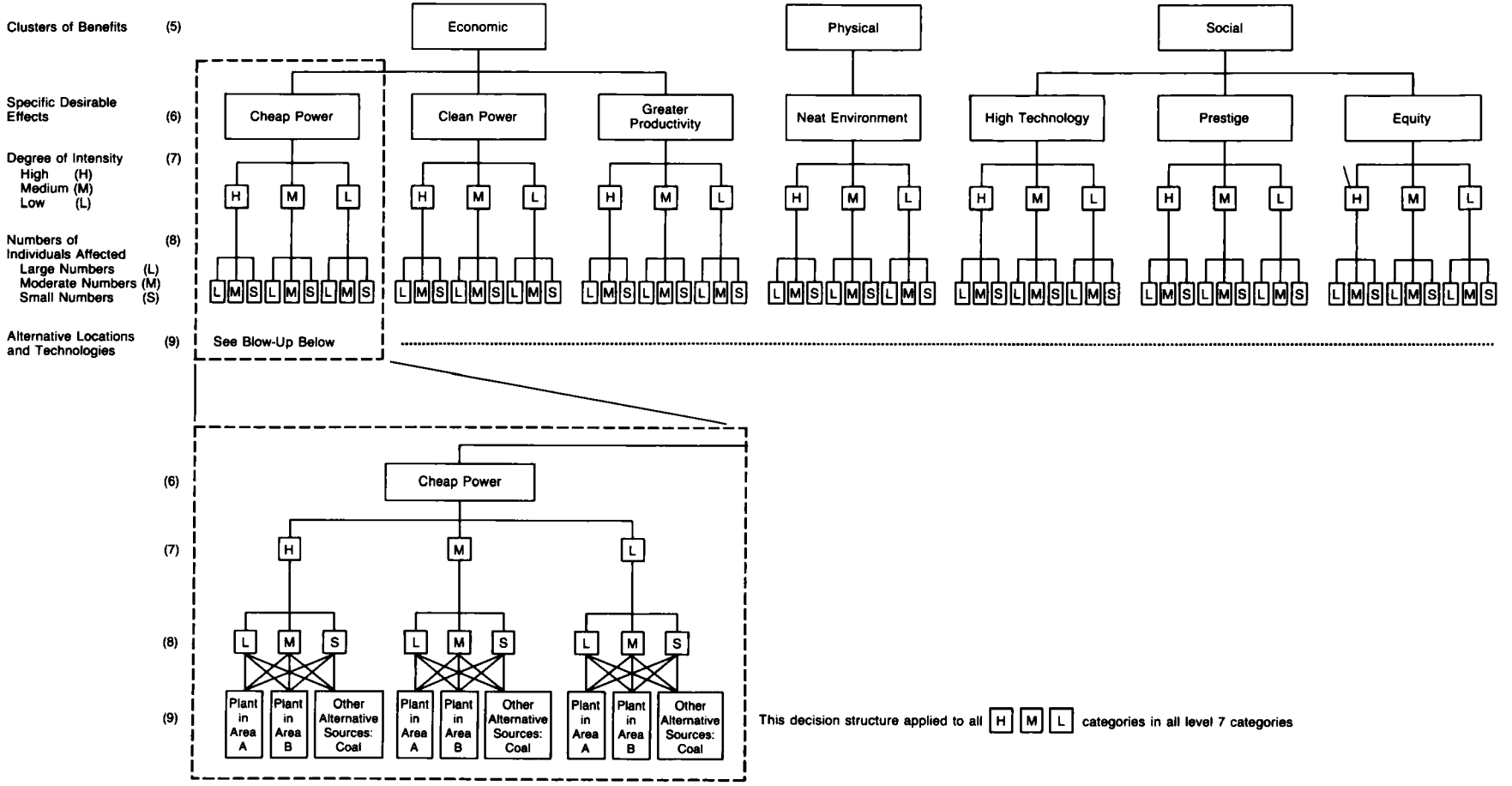

Fig. 3. Benefits hierarchy for siting of a nuclear power facility.

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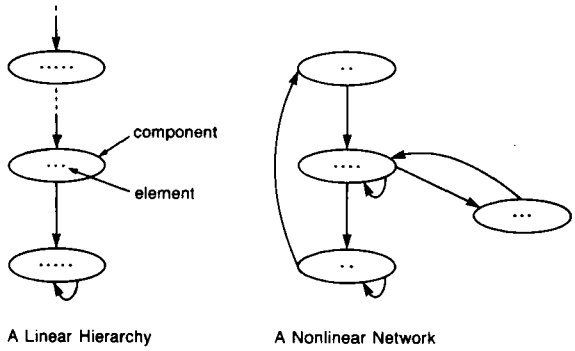

Fig. 4. $A \rightarrow B$ means that A dominates B or that B depends on A.

5. THE GENERAL CASE OF DEPENDENCE IN HIERARCHIC DECISION THEORY

A hierarchy is a simple structure used to represent the simplest type of functional (contextual or semantic) dependence of one level or component of a system on another, in a sequential manner. It is also a convenient way to decompose a complex problem in search of cause-effect explanations in steps which form a linear chain. One result of this approach is to assume the functional independence of an upper part, component, or cluster from its lower parts. This often does not imply its structural independence from the lower parts which involve information on the number of elements, their measurements, etc. But there is a more general way to structure a problem involving functional dependence. It allows for feedback between components, and is a network system of which a hierarchy is a special case. In both hierarchies and networks, the elements within each component may also be dependent on each other[3].

Figure 4 shows two drawings which depict the structural difference between the two frameworks.

In the above figure, a loop means that there is inner dependence of elements within a component.

A nonlinear network can be used to identify relationships among components using one's own

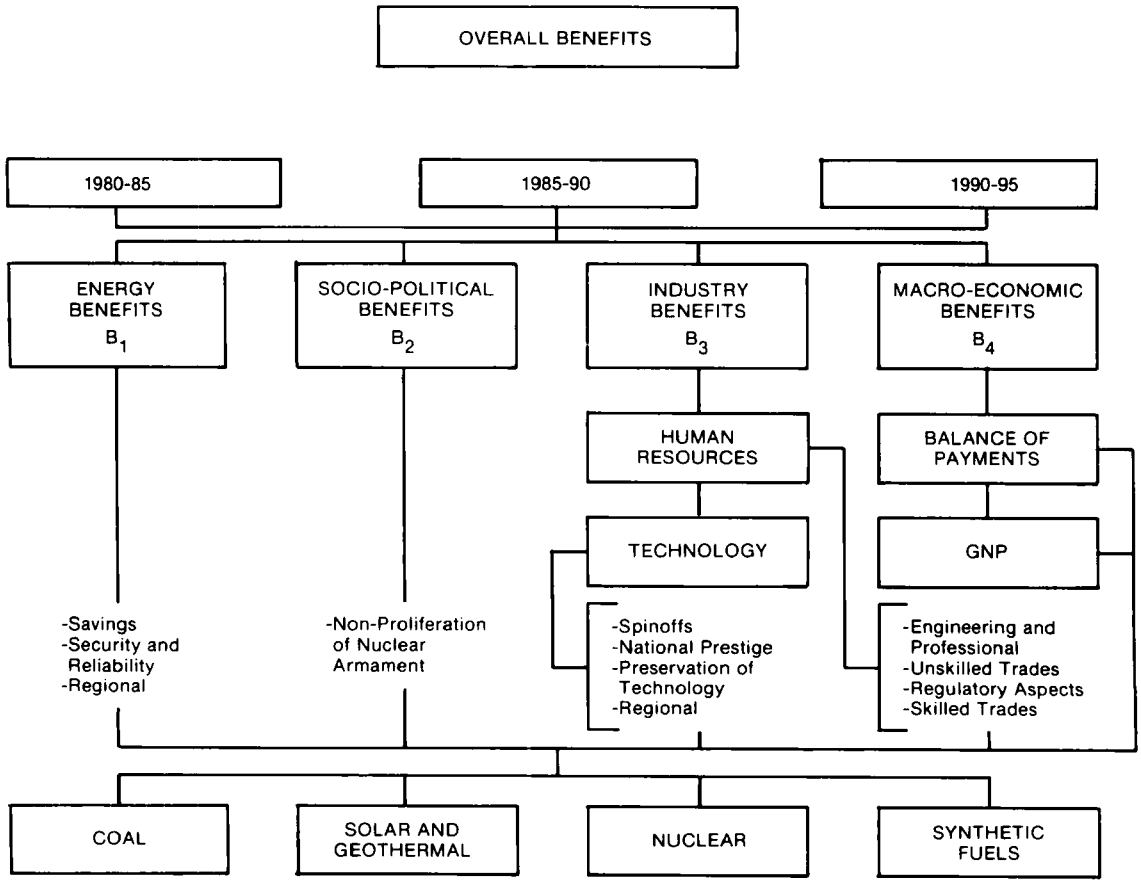

Fig. 5. Benefits hierarchy for power generation alternatives.

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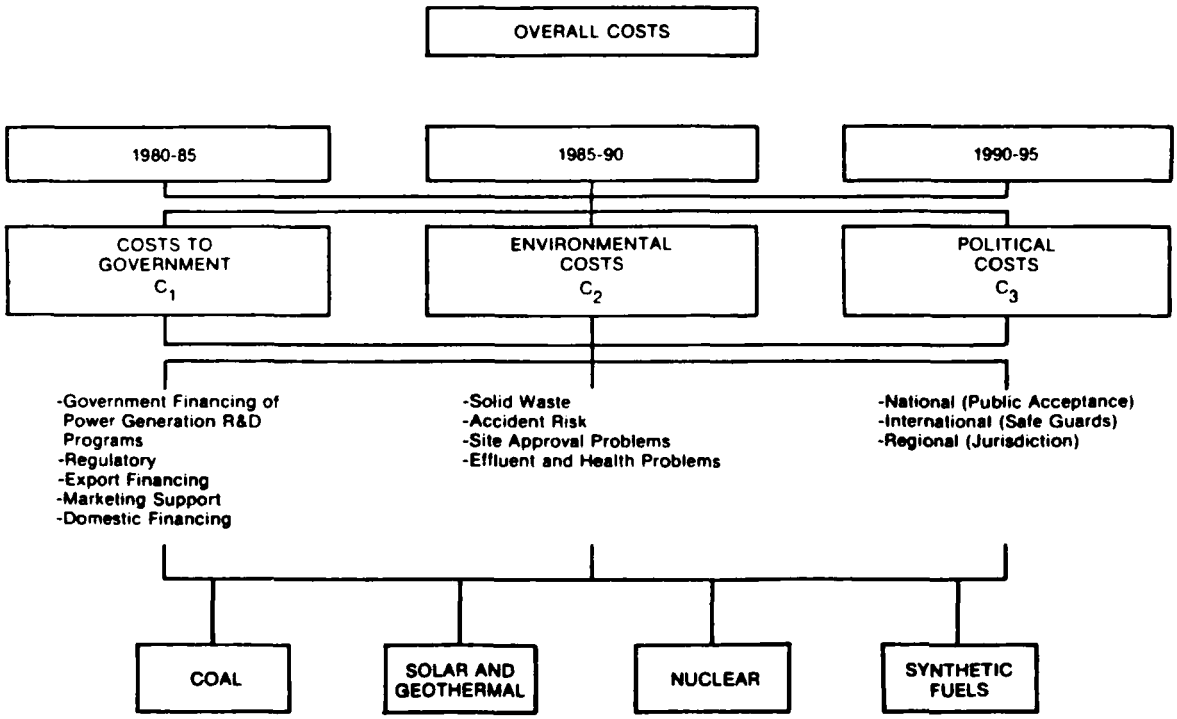

Fig. 6. Costs hierarchy for power generation alternatives.

thoughts, relatively free of rules. It is especially suited for modeling dependence relations. Such a network approach makes it possible to represent and analyze interactions and also to synthesize their mutual effects by a single logical procedure.

For emphasis, we note again that in the nonlinear network diagram or system with feedback (above), there are 2 kinds of dependence: that between components (in a way to allow for feedback circuits), and the interdependence within a component combined with feedback between components. We have called these, respectively, outer and inner dependence.

If the criteria cannot be compared with respect to an overall objective because of lack of information, they can be compared in terms of the alternatives. The systems approach can then be used to replace the hierarchic approach.

Briefly, the system prioritization approach begins with what is known as a supermatrix of blocks of interaction among components. Each column of a block is the eigenvector of priorities of the impact of a component on an element in the system. These eigenvectors are obtained from individual matrices of paired comparisons: one set for comparing criteria in terms of alternatives by answering the question, "Given the alternative, how much more important is one criterion than another for that alternative?"; the other set for comparing alternatives in terms of criteria by answering the question, "Given the criterion, how much more important is one alternative than another for that criterion?"

In this manner, components which depend on one another have impacts which appear in two blocks[3,4]. The overall, or limiting, priorities are obtained from the following supermatrix of interactions. If we denote the 4 criteria by C_1, C_2, C_3, and C_4 and the alternatives by A, B, and C, we have the stochastic supermatrix:

$$W = \begin{array}{c|ccccccc} & C_1 & C_2 & C_3 & C_4 & A & B & C \\ \hline C_1 & 0 & 0 & 0 & 0 & .6279 & .6279 & .6279 \\ C_2 & 0 & 0 & 0 & 0 & .0942 & .0942 & .0942 \\ C_3 & 0 & 0 & 0 & 0 & .2060 & .2060 & .2060 \\ C_4 & 0 & 0 & 0 & 0 & .0719 & .0719 & .0719 \\ A & .250 & .500 & .556 & .545 & 0 & 0 & 0 \\ B & .333 & .333 & .286 & .273 & 0 & 0 & 0 \\ C & .417 & .167 & .158 & .182 & 0 & 0 & 0 \end{array}$$

As we shall see later, this is an irreducible imprimitive matrix. The limiting priorities of the criteria and of the alternatives are obtained by solving the eigenvalue problem $Ww = w$, and normalizing the

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first 4 components of w for the priorities of the criteria and the last 3 for the priorities of the alternatives. This gives, respectively, .6279, .0942, .2060, .0719 and .3578, .3190, .3232. In this case, the same results could have been obtained, respectively, as the first 4 nonzero elements from any of the first 4 columns of:

$$\lim_{k \to \infty} (W^2)^k,$$

and the last 3 nonzero elements from its last 3 columns.

5.1. Questions to Ask

The analytic hierarchy process seeks to elicit judgments from people by asking the appropriate question that would produce the intended answer. By asking the wrong question one would obtain nonsensical results. It remains to determine whether there are right questions to ask, what they are, and how readily people can respond to them. A part of the solution lies in experience obtained using the process. The other part depends on our understanding of the types of problems for which one sets priorities. In general, the situation may be normative or descriptive. In the former case, the question for the pairwise comparison should be formed in terms of what is more preferred or desired in order to satisfy a certain criterion, constraint, property, or scenario. In descriptive situations, the question will seek judgments which identify the degree or extent that two alternatives have a certain property (e.g., of two stones which is heavier and how much; of two roses which is more red and so on). More abstractly, 2 criteria may be compared as to a higher level criterion. Which one is more likely to produce, embody, or fulfill that criterion, or is closer to defining or bringing about the criterion. Generically, we ask which is more "important," meaning a greater possessor of the attribute. These 2 categories of questions relate to 2 types of hierarchies discussed in the AHP literature: the forward (descriptive) and the backward (prescriptive) hierarchies that are often combined through iteration into a process of improving the likely outcome towards the desired outcome.

It should be noted here that in a hierarchy with a single top element or focus, comparison of the second level elements with respect to the focus suggests that one asks "on the average" or "on the whole" which alternative is more "preferred" or more "likely." For other levels, one has specific criteria or elements for making comparisons and need not preface the question with "on the average."

5.2. Hierarchies with Cycles

Power Generation Example with Uncertainty

There are problems in which the future must be factored into the decisions taken in the present. In that case, one lays out different time horizons and different criteria (or scenarios) likely to prevail during one or the other of these time periods. They are essentially a discrete characterization of the situation. In a situation like this, one needs to set priorities on the criteria for each time period. One also needs to set priorities for the time periods for each criterion by entering a judgment as to the time during which the criterion is most likely to prevail. As in the first example, the resulting priority vectors are then entered as columns of a supermatrix, representing the interactions of the two levels of the hierarchy. It has the form:

$$A = \begin{pmatrix} 0 & A_{12} \\ A_{21} & 0 \end{pmatrix}.$$

The columns of the submatrix A_{12} correspond to the priority vectors of the criteria in terms of each time period arranged in the proper order, and the columns of A_{21} correspond to the priority vectors of the time periods in terms of the criteria; the matrix A is column stochastic. Again, without details we give Figs. 5 and 6 to show the benefits and costs hierarchies and a summary of overall benefits and costs of the alternatives, their ratios, and marginal ratios for decision purposes. Here, the nuclear alternative is favored. See Table III for the results.

Table III. Results from the Power Generation Benefits and Costs Hierarchies

	Benefits	Costs	Total benefits/costs	Marginal $\frac{B_i - B_j}{C_i - C_j}$
Coal	.137	.152	.901	.901
Solar and geothermal	.169	.216	.782	—
Nuclear	.288	.210	1.371	2.603
Syn-fuels	.406	.432	.940	1.097

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An interesting application of the feedback concept has been used in the analysis of terrorism. A hierarchy was used whose bottom level of alternatives is linked to its top level of criteria, giving rise to a cyclic hierarchy known as a holarchy. The supermatrix application was essential in deriving the priorities for the courses of action to be followed[5].

5.3. Systems with Feedback

Let us consider the general situation of a system in which components affect each other. It is thus desirable to obtain priorities for the impacts of the elements on all other elements in the system. To do this for each criterion, we must perform pairwise comparisons for each component with respect to the elements in each other component with which it interacts. The resulting eigenvector of each matrix is entered as part of a column of a supermatrix, along whose side and at its top all the elements are listed to obtain a measure of their interaction. Thus, all eigenvectors representing the impact of the elements of a component on one element of another component are arranged in a single column next to their corresponding elements. In this manner, we fill out the entire matrix. Next, all eigenvectors corresponding to a pair of interacting components are weighted by the priorities of the component which creates the impact. These priorities are obtained from a separate pairwise comparison matrix of the components. The judgments of that matrix are entered by answering the question: Given a component of the system, which other component affects it most with respect to the given criterion and how strongly (i.e., Which is the most important component affecting it)? The resulting weighted supermatrix has each column adding to unity. One must make sure that the sum is precisely equal to unity.

We are interested in 2 types of priorities. Those that give the influence or impact of one element on any other element in the system are known as the impact priorities. We are also interested in the absolute priority of any element regardless of which elements it influences. Generally, we seek limiting values of the two kinds of priorities. Calculation of these priorities shows where existing trends might lead if there is no change in preferences which affect the priorities. By experimenting with the process of modifying priorities and noting their limiting trends, we may be able to steer a system towards a more desired outcome.

Now for the formal definitions. The discussion below parallels the theory of Markov Chains as given in Gantmacher[6], and adapted for our purpose. If w_{ij} is the impact priority of the ith element on the jth element in the system, then:

$$w_{ij}^{(1)} = w_{ij}$$

$$w_{ij}^{(2)} = \sum_m w_{im} w_{mj}$$

$$w_{ij}^{(h+k)} = \sum_m w_{im} w_{mj}^{(k)}$$

$$w_{ij}^{(h+k)} = \sum_m w_{im}^{(h)} w_{mj}^{(k)}.$$

The sum of the impact priorities along all possible paths from a given element gives the priorities of an element. This amounts to raising the matrix W to powers. The last expression is equivalent to $W^{(h+k)} = W^h W^k$.

Given that the initial priority of the ith element is $w_i^{(0)}$, we have the following absolute priority of the jth element in paths of length $k \neq 0$:

$$w_j^{(k)} = \sum_i w_i^{(0)} w_{ij}^{(k)}.$$

The problem is to find the limiting impact priority (LIP) matrix W^∞ and the limiting absolute priority (LAP) vector w^∞, as $k \to \infty$. For a priority system, we may also be interested in determining priorities for finite values of k (that does not present problems of existence as does the limiting case). Of particular interest, is to determine when the LAP priority is independent of the initial priorities $w_i^{(0)}$. Such independence is called the ergodicity of the system.

The following is a classification of elements useful in characterizing a system. The reader may wish to go on to the actual discussion of existence and construction of LIP and LAP solutions.

The element j can be reached from the element i if for some integer $k \geq 1$, $w_{ij}^{(k)} > 0$ where $W^k = (w_{ij}^{(k)})$. Here W^k gives the k-reach of each element. A subset of elements C of a system is closed (opposite definition to that for Markov chains) if $w_{ij}^{(k)} = 0$ whenever i is in C and j is not in C. It follows that no element can be reached from any element not in C. The subset C is minimal if it contains no proper closed subset of elements. A set of elements which forms a minimal closed subset corresponds to what is

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known as an irreducible matrix; the system or subsystem itself is called irreducible. A system is called decomposable if it has 2 or more closed sets.

If we initially start with the jth element for some fixed j and denote its first impact on itself in a path of length $k \geq 1$ by $f_j^{(k)}$, we have $f_j^{(1)} = w_{jj}^{(1)}$, $f_j^{(2)} = w_{jj}^{(2)} - f_j^{(1)} w_{jj}^{(1)} \ldots f_j^{(k)} = w_{jj}^{(k)} - f_j^{(1)} w_{jj}^{(k-1)} - \cdots - f_j^{(k-1)} w_{jj}$ and $f_j^{\infty} = \sum_{k=1}^{\infty} f_j^{(k)}$ gives the cumulative impact of j on itself. The mean impact (of j on itself) is given by $u_j = \sum_{k=0}^{\infty} k f_j^{(k)}$.

According to priority influence, we have (the new terms introduced below are essential, as we are not dealing with time transitions):

1. If $f_j = 1$, j is called an enduring (recurrent) element. Thus, an element is enduring if the sum of its impact priorities on itself in a single step (by a loop), in two steps (through a cycle involving one other element), in three steps (involving two other elements, etc.) is equal to unity.

2. If $f_j > 1$, j is called transitory (transient). An element j that is either enduring or transitory is called cyclic (periodic) with cyclicity c, if u_j has values c, $2c$, $3c$, ... where c is the greatest integer greater than unity with this property ($w_{ij}^{(k)} = 0$ where k is not divisible by c). An enduring element j for which u_j is infinite is called fading (null). An enduring element j that is neither cyclic nor fading (i.e., $u_j > \infty$) is called sustaining (ergodic). For either a transitory or a fading element j, $w_{ij}^{(k)} \to 0$ for every i.

If one element in an irreducible subsystem is cyclic with cyclicity c, all the elements in that subsystem are cyclic with cyclicity c. It is known that if j is a sustaining element, then as $k \to \infty$, $w_{jj}^{(k)} \to 1/u_j$; j is a fading element if this number is zero and sustaining if it is positive. Either all the elements of an irreducible subsystem are all transitory or all enduring, and the system itself is called transitory or enduring, respectively.

Remark. The following expression always exists whether a system is irreducible or not. In the former case, its values are known and are as indicated:

$$\lim_{m \to \infty} \sum_{k=0}^{m-1} w_{ij}^{(k)} = \begin{cases} 0 \text{ if } i \text{ and } j \text{ are transitory} \\ 1/u \text{ if } i \text{ and } j \text{ are enduring} \end{cases}.$$

All finite systems of elements must have at least one sustaining element which generates a closed irreducible subset of elements. Since the enduring elements of a finite system are all sustaining, the block (or component) thus generated is called sustaining.

If j is cyclic with cyclicity $c > 1$ then $w_{jj}^{(k)} = 0$ if k is not a multiple of c and $w_{jj}^{(m)} \to c/u$ as $m \to \infty$; $k = mc$, m positive and c the largest integer for which $k = mc$ holds.

We had said earlier that reducibility and primitivity play an important role in proving the existence of LIP and LAP. We now give a few basic facts relating these concepts which will be useful in the ensuing discussion.

A nonnegative irreducible matrix is primitive if it has a unique principal eigenvalue. If the matrix has another eigenvalue with the same modulus as the principal eigenvalue, it is called imprimitive. If the principal eigenvalue has multiplicity greater than unity (equal to unity), but there are not other eigenvalues of the same modulus as the principal eigenvalue, then the matrix is called proper (regular).

A primitive matrix is always regular and, hence, proper but not conversely (e.g., the identity matrix which has unity as an eigenvalue of multiplicity equal to the order of the matrix). A matrix is proper if, and only if, in the normal form, the isolated blocks are primitive. For a regular matrix, the number of isolated blocks is unity.

We note that if all the entries of W are positive, we have a primitive matrix and the theorem on stochastic primitive matrices applies; both LIP and LAP exist. LIP and LAP are the same and are given by the solution of the eigenvalue problem: $Ww = w$. Actually w is any column of $\lim_{k \to \infty} W^k$. The same result is true if W is a primitive matrix.

In general, the nonnegative matrix W may have some zeros. In that case, it is either an irreducible or a reducible matrix. If it is irreducible, then it is either primitive, in which case the above discussion applies, or it is imprimitive. In the latter case, it has a number c of eigenvalues (called the index of imprimitivity) that are not equal to unity whose moduli are equal to unity. This number plays an important role in the solution of the general case from which we can also obtain the solution to this case. It is sufficient to point out here that $W, W^2, \ldots, W^{c-1}$ are all not proper and multiples of these matrices tend toward periodic repetition. The system is cyclic with cyclicity c.

Remark. The system is acyclic, cyclic, irreducible, or reducible, depending on whether the corresponding matrix W is primitive, imprimitive, irreducible, or reducible.

If W is nonnegative and reducible, then it is reduced to the normal form. If the isolated blocks are

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primitive, they are said to correspond to unessential components. The system is by definition called proper and LIP and LAP exist[6].

Important Remark. When our column stochastic matrix is reducible, its essential components drive the system since they are "sources" or impact-priority-diffusing components, as opposed to "sinks" or transition-probability-absorbing states of a Markov chain. In any diagram, except for loops, arrows initiate from and nonterminate at such components.

The solution for LIP is given by:

$$W^{\infty} = \lim_{k \to \infty} W^k = \frac{(I-W)^{-1}\psi(1)}{\psi'(1)},$$

where $\psi(\lambda)$ is the minimum polynomial of W and $\psi'(\lambda)$ is its first derivative with respect to λ. Each column W is a characteristic vector of W corresponding to $\lambda_{\max} = 1$. If $\lambda_{\max} = 1$ is simple (i.e., W is regular), $\psi(\lambda)$ may be replaced by $\Delta(\lambda)$, the characteristic polynomial of W. LAP is obtained as $w^{\infty} = W^{\infty} w^{(0)}$ if W is proper, and the eigenvector solution of $Ww^{\infty} = w^{\infty}$ if W is regular.

Remark. One can show that the matrices of W corresponding to essential components are positive, and those to priority impacts from essential to unessential components are also positive. Only impacts from unessential to unessential or from unessential to essential components are zero.

Finally, if not all isolated blocks are primitive, then each has an index of imprimitivity, as we pointed out earlier. We consider the least common multiple of these which is the cyclicity c of the system. Using the powers of W, LIP is given by:

$$\tilde{W} = (1/c)(I + W + \cdots W^{c-1})(W^c)^{\infty}$$

$$= (1/c)(I - W^c)(1 - W)^{-1}(W^c)^{\infty},$$

and LAP is given by $w = \tilde{W} w^{(0)}$. Both $\tilde{W}$ and w are called the mean LIP and mean LAP, respectively. If there is a single isolated block, then the mean LAP are independent of the initial priorities and are uniquely determined by the solution of: $Ww = w$.

This is precisely the case of an irreducible imprimitive system. Several applications have been made to calculate priorities in a system with feedback. The calculations are long, but the foregoing theory has been found very useful for the purpose. Let us note in closing that the supermatrix of a hierarchy has the following form:

$$W = \begin{bmatrix} 0 & 0 & \cdot & \cdot & \cdot & 0 & 0 & 0 \\ W_{21} & 0 & \cdot & \cdot & \cdot & 0 & 0 & 0 \\ 0 & W_{32} & \cdot & \cdot & \cdot & 0 & 0 & 0 \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ & & & & & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & W_{n-1,n-2} & 0 & 0 \\ 0 & 0 & \cdot & \cdot & \cdot & 0 & W_{n,n-1} & I \end{bmatrix}$$

This matrix has the stable form:

$$W^k = \begin{bmatrix} 0 & 0 & \cdots & 0 & 0 & 0 \\ 0 & 0 & \cdots & 0 & 0 & 0 \\ \cdot & \cdot & \cdots & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdots & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdots & \cdot & \cdot & \cdot \\ 0 & 0 & \cdots & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} W_{n,n-1}W_{n-1,n-2} \cdots W_{32}W_{21}, W_{n,n-1} \\ \cdots W_{32}, \cdots W_{n,n-1}W_{n-1,n-2}, W_{n,n-1} \end{bmatrix}$$

for all $k \geq n-1$. Each coefficient in the last row gives the composite priority impact of the last component on each of the remaining components. Note that the principle of hierarchical composition appears in the $(n,1)$ position as the impact of the nth component on the first. The nth component drives the hierarchy and is the counterpart of an absorbing state in a Markov chain. It is a component of elements which diffuse or are a source of priority impacts. The essence of the above is summarized by the Principle of Hierarchical Composition: the composite vector of a hierarchy of n levels is the entry in the $(n,1)$ position of W^{k-1}, $k \geq n-1$.

This discussion shows that the composition process in a hierarchy, which is additive, conforms with the general composition process of a system with feedback obtained by an alternative approach well-known in classical mathematics.

6. CONCLUSIONS

The foregoing ideas have been applied in a variety of situations involving risk and uncertainty in investment and in corporate and governmental planning, including a study on terrorism. It is generally useful in such cases to carry out the evaluation over a period of time with different groups and synthesize the outcomes according to the group procedure of the Analytic Hierarchy Process. The reader may consult the many references cited[7,8] for further ideas on the AHP and its applications.

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